MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website: mathsmanthan.co.in

Subject: Mathematics | **PERIODIC TEST – 03 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks:** 30

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

Two dice are rolled one after the other. The probability that the number on the first die is strictly smaller than the number on the second die is:

(a) $\frac{15}{36}$ (b) $\frac{5}{12}$
(c) $\frac{1}{2}$ (d) **Both (a) and (b)**

Working & Calculation:

- Total elementary outcomes on rolling two dice: $n(S) = 6 \times 6 = 36$.
- Outcomes with equal numbers: $\{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\} \implies 6$ pairs.
- Unequal outcomes: $36 - 6 = 30$. By symmetry, exactly half have first die $<$ second die:

$$n(E) = \frac{30}{2} = 15 \implies P(\text{first die } < \text{ second die}) = \frac{15}{36} = \frac{5}{12}$$

- Since $\frac{15}{36}$ (Option a) and $\frac{5}{12}$ (Option b) represent the exact same probability, both are correct.

CORRECT ANSWER: OPTION (d) ⟶ Both (a) and (b) $\left[P = \frac{15}{36} = \frac{5}{12}\right]$

Question 2 [1 Mark]

All kings and queens are removed from a pack of 52 cards. A card is drawn from the remaining cards. The probability of getting a black face card is:

(a) $\frac{1}{22}$ (b) $\frac{2}{44}$
(c) $\frac{1}{44}$ (d) **Both (a) and (b)**

Working & Calculation:

- Cards removed: 4 Kings + 4 Queens = 8 cards. Remaining cards: $n(S') = 52 - 8 = 44$ cards.
- Remaining face cards: Originally 12 face cards (4 Kings, 4 Queens, 4 Jacks). After removing Kings and Queens, only the 4 Jacks remain.
- Favourable black face cards are the 2 black Jacks (Jack of Spades, Jack of Clubs) $\implies n(E) = 2$.

$$P(\text{black face card}) = \frac{n(E)}{n(S')} = \frac{2}{44} = \frac{1}{22}$$

- Both $\frac{1}{22}$ (Option a) and $\frac{2}{44}$ (Option b) are numerically identical.

CORRECT ANSWER: OPTION (d) ⟶ Both (a) and (b) $\left[P = \frac{2}{44} = \frac{1}{22}\right]$

Question 3 [1 Mark]

A jar contains 54 marbles each of which is blue, green, or white. The probability of selecting a blue marble is $\frac{1}{3}$ and that of selecting a green marble is $\frac{4}{9}$. How many white marbles does the jar contain?

(a) 12 **(b) 18**
(c) 24 **(d) 6**

Working & Calculation:

- Total marbles $n(S) = 54$.
- Blue marbles $= \frac{1}{3} \times 54 = 18$; Green marbles $= \frac{4}{9} \times 54 = 24$.
- White marbles $= 54 - (18 + 24) = 54 - 42 = 12$ marbles.
- Via complement: $P(\text{white}) = 1 - \left(\frac{1}{3} + \frac{4}{9}\right) = 1 - \frac{7}{9} = \frac{2}{9} \implies \text{Count} = \frac{2}{9} \times 54 = 12$.

CORRECT ANSWER: OPTION (a) $\longrightarrow$ 12 white marbles

Question 4 [1 Mark]

Cards numbered from 6 to 50 are put in a box. The probability of drawing a card with a prime number less than 20 is:

(a) $\frac{4}{45}$ **(b) $\frac{5}{45}$**
(c) $\frac{1}{9}$ **(d) Both (a) and (c)**

Working & Calculation:

- Total cards in the box: $n(S) = 50 - 6 + 1 = 45$ cards.
- Primes in $\{6, 7, \ldots, 50\}$ strictly less than 20: $E = \{7, 11, 13, 17, 19\} \implies n(E) = 5$ cards.
- Probability: $P(\text{prime } < 20) = \frac{n(E)}{n(S)} = \frac{5}{45} = \frac{1}{9}$.
- Both Option (b) $\left[\frac{5}{45}\right]$ and Option (c) $\left[\frac{1}{9}\right]$ are mathematically equal and valid.

CORRECT ANSWER: OPTION (b) / (c) $\longrightarrow$ $P = \frac{5}{45} = \frac{1}{9}$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): The probability that a leap year has 53 Sundays is $\frac{2}{7}$.

Reason (R): A leap year has 366 days, which consists of 52 complete weeks and 2 extra days that can be any of the 7 consecutive day pairs.

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

Step-by-Step Logical Analysis:

1. **Reason (R):** 366 days = 52 weeks + 2 extra days. The 7 consecutive pairs are (Sun, Mon), (Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat), (Sat, Sun). $\implies$ **(R) is TRUE.**
2. **Assertion (A):** Favourable pairs containing Sunday: $\{(\text{Sat, Sun}), (\text{Sun, Mon})\} \implies$ 2 pairs. Hence, $P(53 \text{ Sundays}) = \frac{2}{7}$. $\implies$ **(A) is TRUE.**
3. **Explanation Check:** The 2 extra days detailed in (R) directly establish the favourable fraction $\frac{2}{7}$. Hence, (R) correctly explains (A).

CORRECT ANSWER: OPTION (a) $\longrightarrow$ Both (A) and (R) are true, and (R) is the correct explanation of (A).

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

Question 6 [2 Marks]

A card is drawn from a well-shuffled pack of 52 cards. Find the probability of getting: (i) a non-face card (ii) a red ten.

Sample Space: Total cards in a standard deck $n(S) = 52$.

(i) A non-face card [1 Mark]

Total face cards (Kings, Queens, Jacks) $= 3 \times 4 = 12$. Non-face cards $= 52 - 12 = 40$:

$$P(\text{non-face card}) = \frac{40}{52} = \frac{10}{13}$$

Answer (i): $\mathbf{P(\text{non-face card}) = \frac{10}{13}}$

(ii) A red ten [1 Mark]

Red suits are Hearts and Diamonds. Favourable cards are 10 of Hearts and 10 of Diamonds ($n(E) = 2$):

$$P(\text{red ten}) = \frac{2}{52} = \frac{1}{26}$$

Answer (ii): $\mathbf{P(\text{red ten}) = \frac{1}{26}}$

Question 7 [2 Marks]

It is given that in a group of 3 students, the probability of 2 students not having the same birthday is 0.992. What is the probability that the 2 students have the same birthday?

Step-by-Step Derivation:

1. Let event E be "the two students have the same birthday".
2. Its complement $\bar{E}$ represents "the two students do not have the same birthday".
3. Given $P(\bar{E}) = 0.992$. Since $P(E) + P(\bar{E}) = 1$:

$$P(E) = 1 - P(\bar{E}) = 1 - 0.992 = 0.008 = \frac{8}{1000} = \frac{1}{125}$$

Answer: $\mathbf{P(\text{two students have same birthday}) = 0.008 = \frac{1}{125}}$

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

Question 8 [3 Marks]

A die has its six faces marked 0, 1, 1, 1, 6, 6. Two such dice are thrown together and the total score is recorded. (i) List all the possible distinct values of the total score. (ii) Find the probability of getting a total of 7. (iii) Find the probability of getting a total score of at least 2.

Outcomes Breakdown (36 total): Die faces: 0 (1 face), 1 (3 faces), 6 (2 faces).

- Sum 0: $1 \times 1 = 1$
- Sum 1: $1 \times 3 + 3 \times 1 = 6$
- Sum 2: $3 \times 3 = 9$
- Sum 6: $1 \times 2 + 2 \times 1 = 4$
- Sum 7: $3 \times 2 + 2 \times 3 = 12$
- Sum 12: $2 \times 2 = 4$.

(i) All possible distinct values of the total score [1 Mark]

The distinct total scores obtained are: $\{0, 1, 2, 6, 7, 12\}$.

Answer (i): **Distinct total scores:** $\mathbf{\{0, 1, 2, 6, 7, 12\}}$

(ii) Probability of getting a total of 7 [1 Mark]

Score 7 occurs via $(1, 6)$ [6 ways] and $(6, 1)$ [6 ways] $\implies n(E_2) = 12 \implies P = \frac{12}{36} = \frac{1}{3}$.

Answer (ii): $\mathbf{P(\text{total of 7}) = \frac{12}{36} = \frac{1}{3}}$

(iii) Probability of getting a total score of at least 2 [1 Mark]

Scores < 2 are 0 (1 way) and 1 (6 ways) $\implies$ 7 ways. Favourable scores ≥ 2: $36 - 7 = 29 \implies P = \frac{29}{36}$.

Answer (iii): $\mathbf{P(\text{total} \geq 2) = \frac{29}{36}}$

Question 9 [3 Marks]

A carton consists of 100 shirts of which 88 are good, 8 have minor defects, and 4 have major defects. Jimmy, a trader, will only accept shirts which are good, but Sujatha, another trader, will only reject shirts which have major defects. One shirt is drawn at random from the carton. What is the probability that: (i) it is acceptable to Jimmy? (ii) it is acceptable to Sujatha? (iii) it has major defects?

Given Data: Total shirts $n(S) = 100$. Good $= 88$, Minor defects $= 8$, Major defects $= 4$.

(i) Acceptable to Jimmy: Jimmy accepts only good shirts $(n(E_1) = 88) \implies P = \dfrac{88}{100} = \dfrac{22}{25} = 0.88$.

Answer (i): $\mathbf{P(\text{acceptable to Jimmy}) = \dfrac{88}{100} = \dfrac{22}{25} = 0.88}$

(ii) Acceptable to Sujatha: Sujatha accepts good and minor-defect shirts $(88 + 8 = 96) \implies P = \dfrac{96}{100} = \dfrac{24}{25} = 0.96$.

Answer (ii): $\mathbf{P(\text{acceptable to Sujatha}) = \dfrac{96}{100} = \dfrac{24}{25} = 0.96}$

(iii) Shirt has major defects: Number of shirts with major defects is $4 \implies P = \dfrac{4}{100} = \dfrac{1}{25} = 0.04$.

Answer (iii): $\mathbf{P(\text{major defects}) = \dfrac{4}{100} = \dfrac{1}{25} = 0.04}$

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [5 Marks]

Question 10 [5 Marks]

A box contains 90 discs numbered from 1 to 90. If one disc is drawn at random from the box, calculate the probability that it bears:
(i) a two-digit number (ii) a perfect square number (iii) a prime number less than 30
(iv) a number divisible by both 3 and 5 (v) a number divisible by 2 or 3.

Sample Space: Discs are numbered $1, 2, 3, \ldots, 90 \implies n(S) = 90$.

(i) A two-digit number: Single-digit numbers 1 to 9 (9 discs) $\implies$ Two-digit discs $= 90 - 9 = 81 \implies P = \dfrac{81}{90} = \dfrac{9}{10} = 0.9$.

Answer (i): $\mathbf{P(\text{two-digit number}) = \dfrac{9}{10} = 0.9}$

(ii) A perfect square number: Squares in 1 to 90 are $\{1, 4, 9, 16, 25, 36, 49, 64, 81\}$ (9 numbers) $\implies P = \dfrac{9}{90} = \dfrac{1}{10} = 0.1$.

Answer (ii): $\mathbf{P(\text{perfect square}) = \dfrac{1}{10} = 0.1}$

(iii) A prime number less than 30: Primes are $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$ (10 numbers) $\implies P = \dfrac{10}{90} = \dfrac{1}{9}$.

Answer (iii): $\mathbf{P(\text{prime } < 30) = \dfrac{1}{9}}$

(iv) Divisible by both 3 and 5: Multiples of LCM$(3, 5) = 15$: $\{15, 30, 45, 60, 75, 90\}$ (6 discs) $\implies P = \dfrac{6}{90} = \dfrac{1}{15}$.

Answer (iv): $\mathbf{P(\text{divisible by both 3 and 5}) = \dfrac{1}{15}}$

(v) Divisible by 2 or 3: Multiples of 2: $\lfloor 90/2 \rfloor = 45$; multiples of 3: $\lfloor 90/3 \rfloor = 30$; multiples of 6: $\lfloor 90/6 \rfloor = 15$.

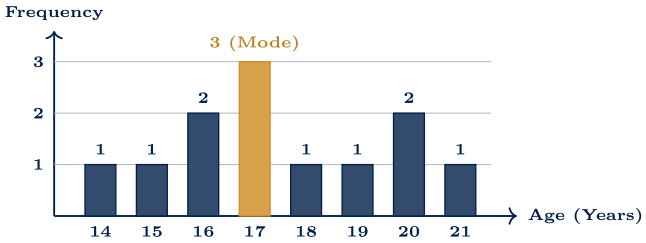
By Principle of Inclusion-Exclusion: $n(2 \text{ or } 3) = 45 + 30 - 15 = 60 \implies P = \dfrac{60}{90} = \dfrac{2}{3}$.

Answer (v): $\mathbf{P(\text{divisible by 2 or 3}) = \dfrac{2}{3}}$

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 11 (Case Study Analysis) [4 Marks]

A social science survey recorded the ages (in completed years) of 12 senior secondary students selected for a youth council:
14, 17, 15, 21, 17, 20, 16, 18, 17, 16, 19, 20
One student is selected at random from this council. Based on the above scenario, answer the following:



Ordered Data & Sample Space: Total council members $n(S) = 12$.
Arranged in ascending order: $14, 15, 16, 16, 17, 17, 17, 18, 19, 20, 20, 21$.

(i) Age 18 or older (eligible to vote) [1 Mark]

Ages ≥ 18: $\{18, 19, 20, 20, 21\} \implies n(E_1) = 5$ students $\implies P = \dfrac{5}{12}$.

Answer (i): $\mathbf{P(\text{age } \geq 18) = \dfrac{5}{12}}$

(ii) Strictly younger than 16 years [1 Mark]

Ages < 16: $\{14, 15\} \implies n(E_2) = 2$ students $\implies P = \dfrac{2}{12} = \dfrac{1}{6}$.

Answer (ii): $\mathbf{P(\text{age } < 16) = \dfrac{2}{12} = \dfrac{1}{6}}$

(iii) Mode of the age distribution and probability of modal age [1 Mark]

The value with the greatest frequency (3 times) is 17 $\implies$ Mode = 17 years.

$$P(\text{modal age}) = \frac{3}{12} = \frac{1}{4} = 0.25$$

Answer (iii): $\mathbf{\text{Mode} = 17 \text{ years}, \quad P(\text{modal age}) = \dfrac{1}{4} = 0.25}$

(iv) Probability that student's age is a prime number [1 Mark]

Prime ages are 17 (3 students) and 19 (1 student) $\implies n(E_4) = 3 + 1 = 4 \implies P = \dfrac{4}{12} = \dfrac{1}{3}$.

Answer (iv): $\mathbf{P(\text{prime age}) = \dfrac{4}{12} = \dfrac{1}{3}}$

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